Self-Dual BKZ on NTRU: Dimension-Onset, Fatigue, and a Cross-Engine Reference-Free DSD Observable

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Abstract

This technical report extends the self-dual BKZ (SD-BKZ) reduction benchmark of the companion paper from LWE-Kannan lattices to *NTRU*, and from a single enumeration engine (fplll) to a second, sieving engine (G6K). We report three empirical results. First, on circulant NTRU lattices the SD-BKZ advantage exhibits a sharp *dimension-onset*: tied with BKZ for small n, a high-variance spike around $n \approx 71$–73 (mean advantage up to $+9.7$ nats), then a tight $+1.5$-nat plateau for $n \geq 79$. Second, the defensible NTRU observable is *reference-free*: signed $d(\mathrm{LN})$ is a reference artifact on NTRU (no canonical fixed point), but the BKZ-vs-SD-BKZ profile divergence / dense-sublattice-discovery (DSD) onset is robust, and the SD-vs-BKZ DSD-onset *gap grows with dimension* ($0\% \to 27\%$ over $n = 67$–113) as q crosses the fatigue point. Third, the cross-engine study confirms the companion paper's thesis from the other side—the Root Hermite Factor is blind to the SD-BKZ difference while $d(\mathrm{LN})$/DSD sees it—now on a sieve oracle, and finds the effect to be regime-dependent. We also re-validate, on a new modulus family, an fplll GSO numerical fix. All experiments are reproducible via a containerised, determinism-gated benchmark.

Keywords: lattice reduction, BKZ, SD-BKZ, NTRU, overstretched, fatigue, dense sublattice discovery, Rankin profile, sieving, G6K

1 Introduction

The companion paper [CB26] showed, on LWE-Kannan lattices, that the Root Hermite Factor (RHF) is structurally blind to differences between BKZ and self-dual BKZ [MW16] that the Li–Nguyen Rankin profile distance $d(\mathrm{LN})$ [LN20] makes visible. This report asks two follow-up questions a data paper is well suited to answer: *(i) does the SD-BKZ phenomenology generalise beyond LWE to NTRU?* and *(ii) does it reproduce on a fundamentally different reduction oracle—sieving (G6K) rather than enumeration?*

Our contributions are primarily empirical and infrastructural:

- A characterisation of the SD-BKZ advantage on circulant NTRU lattices [DvW21], including a sharp dimension-onset transition (Section 4).
- Evidence that the appropriate NTRU observable is reference-free, and a *dense-sublattice-discovery (DSD) onset* whose SD-vs-BKZ gap grows with dimension as q approaches the fatigue point (Section 5).

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Date of this document: 2026-06-08.

- A cross-engine study (Section 6): G6K wired as a second engine behind a reproducibility-gated seam, a self-dual pump-and-jump construction validated against fplll's SD-BKZ, and confirmation that RHF-blindness is engine-independent but regime-dependent—consistent with independent observations of RHF-equality under a non-exact oracle [Row24].

- An independent re-validation of an fplll GSO catastrophic-cancellation fix on a new (NTRU) modulus family (Section 7).

This is a *technical report / data paper*: the emphasis is on the experimental apparatus, the data, and reproducibility, rather than on new theory. Closest prior work is Deng and Jia [J+23], who give a sieving-based self-dual BKZ that puts a G6K Pump oracle in place of enumeration and thereby attains comparable or better output quality at substantially lower running time, and who observe the SD-vs-BKZ quality gap narrowing as the block size grows. Our aim is complementary rather than competing: we do not propose a faster oracle but a finer empirical characterisation, measuring the SD-vs-BKZ structural difference through the reference-free profile observable across a fixed-β, q-swept regime. White [Whi26] studies dynamic block-size strategies within the same Li–Nguyen framework, complementary to the fixed-β, q-swept regime characterised here.

2 Background

2.1 Circulant NTRU lattices and fatigue

We use the circulant NTRU construction of Ducas and van Woerden [DvW21]. For a prime n and modulus q, sample ternary secrets $f, g \in \mathbb{Z}_q[x]/(x^n - 1)$ (uniform $\{-1, 0, 1\}$, coordinate variance $\sigma^2 = 2/3$) with f invertible, set $h = g/f \bmod q$, and form the $2n \times 2n$ basis

$$B = \begin{pmatrix} qI_n & 0 \\ H & I_n \end{pmatrix},$$

where H is the circulant matrix of h. The secret (f, g) is an unusually short vector in the row span; recovering it breaks the key. Two structures distinguish B from a random q-ary lattice: a *q-vector head* (rows of qI_n that survive reduction as length-q vectors) and a *dense secret sublattice* spanned by the rotations of (f, g). A BKZ-reduced NTRU profile therefore follows the *ZGSA* (Z-shaped) model—a flat head at $\log q$, then a GSA slope—rather than the single-slope GSA line of a random q-ary lattice.

As q grows at fixed n, the secret sublattice becomes relatively denser and easier to detect: at the *fatigue point* $q_{\text{fat}}(n) \approx 0.004\, n^{2.484}$ [DvW21] the attack transitions from the standard regime (secret found only through full SVP) to the *overstretched* regime, where reduction discovers the dense sublattice (*dense-sublattice discovery*, DSD) well before solving SVP. The geometry of this transition—and how self-dual reduction moves through it—is the object of this report.

2.2 BKZ, SD-BKZ, and the two engines

Block Korkine–Zolotarev (BKZ) reduction repeatedly calls an SVP oracle on length-β projected sublattice windows, sweeping left to right. Self-dual BKZ (SD-BKZ) [MW16] alternates a forward (primal) pass with a backward pass on the reversed dual basis, so each tour reduces the basis from both ends. On LWE-Kannan lattices the companion paper [CB26] found the two produce bases of indistinguishable Root Hermite Factor yet measurably different Rankin profiles; this report asks how that difference behaves on NTRU and across two oracle implementations.

We use two engines. **fplll** supplies an exact enumeration SVP oracle; its self-dual variant (`SD_VARIANT`) is our ground truth. **G6K** [ADH+19] supplies an approximate *sieving* oracle through the pump-and-jump BKZ tour (`pump_n_jump_bkz_tour`); we realise SD-BKZ on it as a primal pump-and-jump pass followed by a dual pass run under G6K's `dual_mode` (operations on the dual basis, bounds reflected about full_n/2). The two oracles differ fundamentally—deterministic enumeration versus randomised sieving—which is what makes a phenomenon that survives both engines credible. Sieving carries a fixed setup cost that makes small β uninformative, so all cross-engine experiments use $\beta = 40$, the smallest block size at which the sieve does non-trivial work; this is also the regime in which related q-ary forgery attacks operate [DEP23].

2.3 The reference-free DSD observable

The companion paper's metric is $d(\mathrm{LN}) = \frac{1}{m}\sum_i |r_i - \rho_i|$, where r_i is the slope-removed Gram–Schmidt log-norm profile and ρ_i is the Li–Nguyen BKZ fixed point [LN20]—a GSA line valid for a *random q-ary* lattice. NTRU is not random q-ary (Section 2.1): its reduced profile follows the ZGSA Z-shape, for which no closed-form Li–Nguyen fixed point exists. Comparing an NTRU profile to the LWE line therefore conflates structure with reduction dynamics, and the *signed* advantage $d(\mathrm{LN})_{\mathrm{BKZ}} - d(\mathrm{LN})_{\mathrm{SD}}$ can flip sign under a different reference—it is a *reference artifact*.

Two observables survive this. First, the *unsigned* profile divergence $\frac{1}{m}\sum_i |r_i^{\mathrm{BKZ}} - r_i^{\mathrm{SD}}|$ is reference-free (it never names ρ); on NTRU it is large precisely at the transition and reproduces under three independent references. Second, and what we report throughout, is a binary *dense-sublattice-discovery* flag: a reduction has found the secret sublattice when $b_1 = \min_i \log \|\mathbf{b}_i^*\| > 1.5$, where the threshold is the q-independent ternary-secret log-norm floor $\log\sqrt{2n \cdot 2/3} \approx 1.96$ (we also track the count of GS vectors that have collapsed below the q-vector head, $\leq n+1$). Both are properties of the output basis alone, independent of any modelled reference. The onset q is the smallest modulus at which a given reduction flags DSD; the SD-vs-BKZ onset *gap* is the object of Sections 5 and 6.

3 Experimental Setup

3.1 Generators and engines

NTRU bases are built by `generators/ntru.py` following Section 2.1; every basis is checked structurally at construction ($\dim = 2n$, a qI_n block, and $Hf \equiv g \pmod q$) before any reduction. Both engines sit behind a single seam: `run_single` takes a `backend` argument, and `make_backend` returns either an fplll or a G6K backend exposing a common `tour()`/`gso()` interface. The per-tour reduction loop is then engine-blind—the same loop drives enumeration and sieving, so the two cannot diverge through duplicated control flow. The G6K self-dual tour is a primal `pump_n_jump_bkz_tour` followed by a dual pass executed inside a `temp_params` block that flips `dual_mode`; this was validated against fplll's `SD_VARIANT` by checking that the SD−BKZ profile-change signature (the GS-slope flattening) agrees in direction and magnitude across engines (fplll -0.00039 vs G6K -0.00040 per tour at $n = 80$, $\beta = 40$).

3.2 Reproducibility and determinism gates

Both engines run in containers built from a digest-pinned base with `snapshot.debian.org` apt pinning and a fixed `-march=x86-64-v3` target, so the binaries are bit-reproducible across machines. Each result is a per-seed JSON carrying the full input parameters and output profile; a SHA-256 manifest gates byte-identity. The fplll path is held to

Table 1: SD-BKZ advantage on NTRU: tied → onset spike → plateau.

n	mean adv	std	seeds>0
59–67	~ 0	tight	3–9/20
71	$+4.35$	5.59	16/20
73	$+9.68$	5.21	19/20
79–89	$+1.5$	~ 0.2	20/20

exact-JSON reproducibility by `verify.sh`; the G6K path by `verify_g6k.sh` against a captured reference hash, confirmed on a second machine (the CI runner).

Determinism on the sieve required care (ADRs 005–008). G6K's multi-threaded sieve has a nondeterministic reduction race, so all sieving runs single-threaded (`threads` $= 1$); across-seed parallelism still uses all cores, since seeds are independent. The fplll global RNG must be seeded both before basis construction and again before the sieve (G6K samples from it), and the `SieverParams` seed is a no-op that we never rely on. With these, multi-tour G6K runs are bit-identical run-to-run, and—verified at $n = 80$, $\beta = 60$, three tours—per-tour re-seeding is itself a no-op: the construction seed alone determines all N tours, so the policy is to seed once at construction. The new modulus and engine introduced no schema change to the seed JSON, preserving the companion paper's SHA baselines.

3.3 Campaigns and parameter grids

Every experiment is a named campaign in `config/sweep.toml`, fixing generator, engine, n-grid, β, q, tour count, and seed count so a single command reproduces a cell. The dimension-onset sweep (Section 4) fixes $q = 97$, $\beta = 20$, 50 tours and sweeps $n \in [59, 89]$ at 20 seeds each. The fatigue and DSD-onset sweeps (Section 5) fix n and sweep q across the fatigue point. The cross-engine sweeps (Section 6) fix $n = 89$, $\beta = 40$, 50 tours and sweep a matched q-grid on *both* engines—the G6K campaign (`ntru_g6k_qsweep`) and the fplll campaign (`ntru_xeng_qsweep`) write to leaves that share the same $(q, \text{precision}, \text{tours}, n, \beta)$ key, so a comparator reads the two engines cell-for-cell. The Kahan re-validation (Section 7) runs $n = 127$ at $q \in \{971, 1087, 1201\}$ through a separate patched-fplll campaign whose seeds land in their own tree, never overwriting the canonical seeds.

4 The SD-BKZ Advantage Has a Dimension-Onset on NTRU

On circulant NTRU at $q = 97$ (standard regime), $\beta = 20$, 50 tours, the SD-BKZ-vs-BKZ advantage has a *sharp* dimension transition, not a smooth ramp (20 seeds per n):

Three regimes: tied for $n \leq 67$; an onset zone at $n = 71$–73 where the mean spikes to $+4$–10 nats with $\sim 25\times$ the plateau variance (SD-BKZ cracks the dense sublattice on most-but-not-all instances); and a stable $+1.5$-nat plateau for $n \geq 79$ (std ≈ 0.2, 20/20 wins). The onset spike exceeds the plateau—non-monotonic (Figure 1).

5 Fatigue and the Cross-Dimension DSD-Onset Gap

Before sweeping q we confirm the metric is sound here. The $n \approx 71$–73 divergence of Section 4 is reference-robust: re-derived from the stored profiles under three independent references (Li–Nguyen, GSA-zero, and the reference-free $\frac{1}{m}\sum_i |r_i^{\mathrm{BKZ}} - r_i^{\mathrm{SD}}|$), it remains large under all three (32.7 nats reference-free at $n = 73$ vs 0.03 at $n = 67$). The

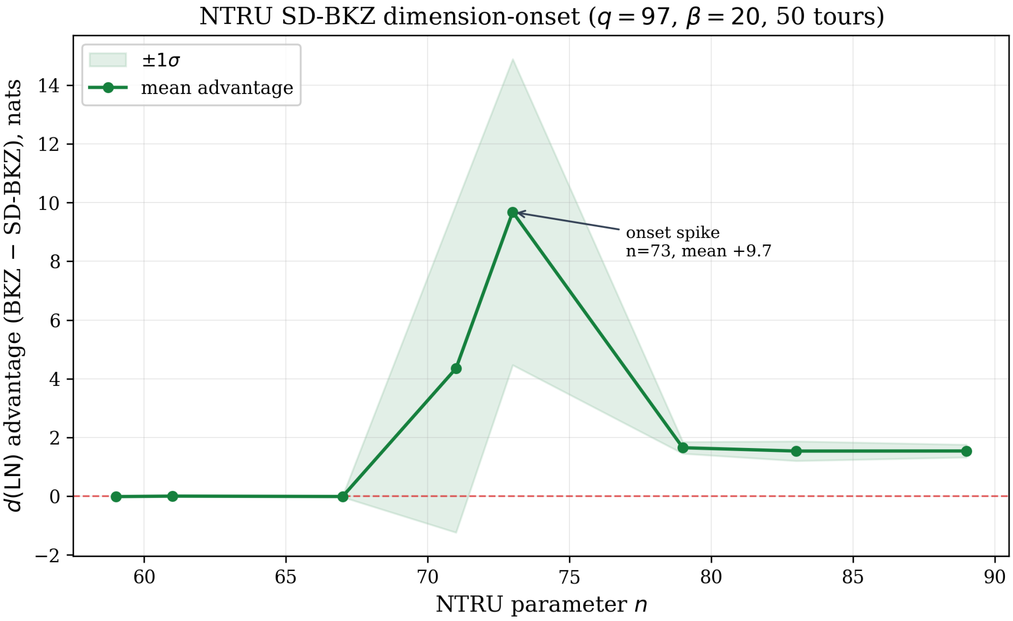

Figure 1: NTRU SD-BKZ dimension-onset ($q = 97$, $\beta = 20$, 50 tours, 20 seeds per n, $\pm 1\sigma$ band). The signed advantage is reference-robust at the spike (Section 2.3); shown here for the dimension transition. Recomputed from the per-seed `advantage` field.

signed advantage, by contrast, is a reference artifact on NTRU (Section 2.3); accordingly everything below reports the reference-free DSD observable, never the signed quantity.

Sweeping q across the fatigue point at fixed n, the reference-free BKZ-vs-SD-BKZ profile divergence peaks near $q \approx q_{\text{fat}}$. The SD-BKZ DSD-onset occurs at progressively *lower* q than the BKZ onset as n grows—the gap widens with dimension:

$n = 127$ is excluded (fplll §8 catastrophic cancellation + an off-grid crack). The widening is shown in Figure 2.

6 Cross-Engine: SD-BKZ on a Sieve Oracle

The G6K engine seam and the self-dual pump-and-jump construction are described in Section 3.1; recall that the construction was validated against fplll's `SD_VARIANT` by

Table 2: Reference-free DSD-onset (proper two-part criterion): SD-BKZ reaches dense-sublattice discovery at lower q than BKZ, the gap growing with n. Seed-backed across the q-sweep (up to 100 seeds/cell), including the $n = 113$ endpoint from the completed precision ladder.

n	SD onset q	BKZ onset q	gap%
67	167	168	1
79	182	183	0
89	238	283	19
101	429	512	20
113	729	925	27

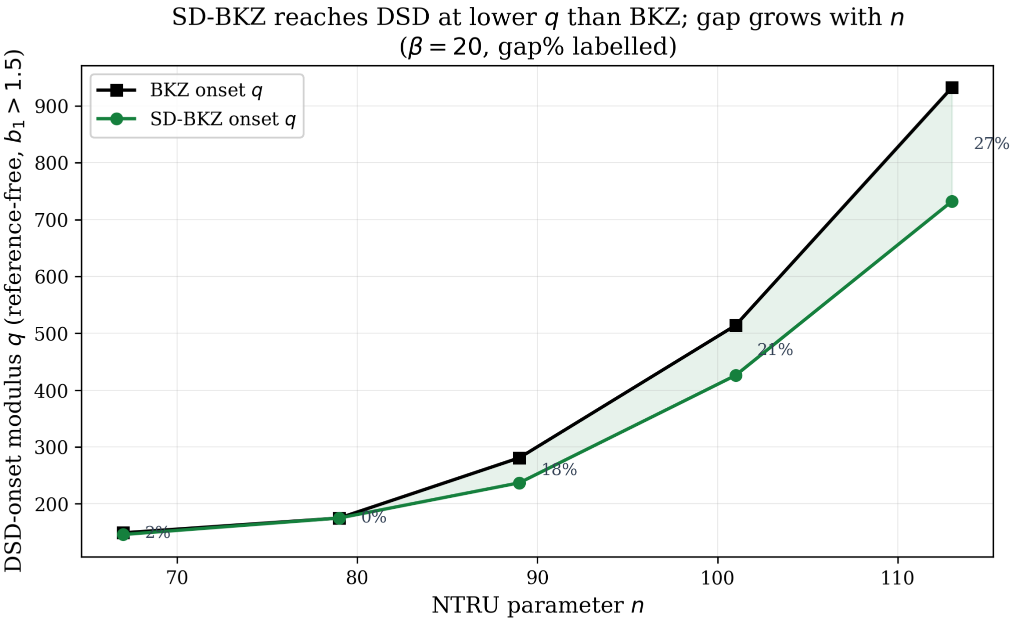

Figure 2: Reference-free DSD-onset modulus ($b_1 > 1.5$) for SD-BKZ vs BKZ as a function of n ($\beta = 20$), with the SD-vs-BKZ gap% labelled. SD-BKZ reaches dense-sublattice discovery at progressively lower q than BKZ; the gap grows from 0 to 27%. Values mirror Table 2.

matching the per-tour GS-slope-flattening signature (-0.00039 vs -0.00040). This section reports what the validated sieve engine then shows.

RHF blindness is engine-independent but regime-dependent. On NTRU at $q = 97$, $\beta = 40$, the companion paper's thesis reproduces on the sieve: in the standard regime the RHF of BKZ and SD-BKZ bases is identical to machine precision while $d(\mathrm{LN})$ differs. The effect is regime-dependent—RHF-blindness breaks only on the rare seeds where SD-BKZ finds a short vector (which RHF reflects). This is consistent with RHF-equality reported under a non-exact (approximate-enumeration) oracle [Row24]; the present work measures the structure RHF misses.

Cross-engine validation of the construction. A dimension-resolved fplll-vs-G6K comparison at $q = 97$ ($n = 89/101/113$, 20 seeds each, $\beta = 40$, 50 tours) confirms the self-dual G6K construction is sound (Table 3): at $n = 101/113$ neither engine fires a short-vector event, and at $n = 89$ the engines overlap (Jaccard 0.50) on the one seed where a short vector genuinely exists (seed 19), where both land the identical first GS log-norm 2.332. The G6K sieve never fires where fplll's exact oracle does not—there is no spurious sieve firing—so the $n = 89$ short-vector events are real but sparse, not an artifact of the approximate oracle.

At a matched $\beta = 40$ the two engines agree: no DSD-onset gap. We run the *same* grid ($n = 89$, $\beta = 40$, 50 tours, $q \in \{97, 113, 137, 157, 181, 211\}$) on both the fplll enumeration engine and the G6K sieve, under the *proper* two-part DSD criterion: a basis has reached dense-sublattice discovery when its profile has collapsed onto the dense sublattice, $\#\{i : \log \|\mathbf{b}_i^*\| < 2.888\} \le n + 1$ *and* $\min_i \log \|\mathbf{b}_i^*\| > 1.5$. (The looser $b_1 > 1.5$

Table 3: Cross-engine soundness at $q = 97$: G6K never fires where fplll does not; the $n = 89$ overlap is the one genuine short-vector seed.

n	G6K fired	fplll fired	Jaccard
89	2/20	1/20	0.50
101	0/20	0/20	1.00
113	0/20	0/20	1.00

Table 4: Cross-engine at $\beta = 40$ under the proper two-part DSD criterion: both engines reach DSD-onset together ($q \approx 196$–211) with no SD-vs-BKZ gap—an honest null. The separate min-GS-clearing event ($\min_i \log \|\mathbf{b}_i^*\| > 1.5$ alone: aggressive tail reduction, *not* DSD) shows SD>BKZ on both engines and a sieve-vs-enum structural difference ($q = 137$, 100 seeds).

	fplll (enum)		G6K (sieve)	
	SD	BKZ	SD	BKZ
two-part DSD @ q=137	0/100	0/100	0/100	0/100
two-part DSD onset	$q \approx 196$–211, both engines, no gap			
min-GS-clearing @ q=137	38/100	12/100	61/100	23/100
median $\min_i \log \|\mathbf{b}_i^*\|$	0.255	0.104	**1.844**	0.140

test alone over-fires on uniformly-reduced sub-fatigue bases, where the tail is cleared but the profile has not collapsed.) Under this criterion the cross-engine picture is a clean *null*: at $q = 137$ neither SD-BKZ nor BKZ fires DSD on *either* engine, and the true DSD onset—$q \approx 196$–211—is reached by SD-BKZ and BKZ *together*, with no SD-vs-BKZ gap on either oracle (Table 4).

The cross-engine result is therefore not a confirmation of the $\beta = 20$ DSD-onset gap of Section 5 but a *regime constraint*: that gap is a $\beta = 20$ phenomenon, while the sieve is only meaningful at $\beta \geq 40$, and the two regimes do not overlap. What the matched-β comparison *does* surface is a structural difference between the oracles themselves. The min-GS-clearing event—an aggressively reduced tail ($\min_i \log \|\mathbf{b}_i^*\| > 1.5$, short of full DSD collapse)—fires more often under SD-BKZ than BKZ on both engines (38 vs 12 on fplll, 61 vs 23 on G6K at $q = 137$, 100 seeds), and the G6K sieve drives the shortest GS log-norm far higher than fplll enumeration on identical instances: median 1.844 (sieve) vs 0.255 (enum), a $\sim 7\times$ gap. The sieve leaves a more aggressively-reduced sub-fatigue tail than enumeration. This is a genuine cross-engine observation—both engines start from the byte-identical primal basis (100/100 seeds) and the determinant is conserved to $\sim 10^{-7}$ (`results/validation/sieve_vs_enum_min_gs_clearing.json`)—but it is aggressive reduction, *not* dense-sublattice discovery, and we label it as such.

The reference-free observable remains essential on these cells: where the min-GS-clearing event fires, the *signed* d(LN) advantage is large and negative on both engines, because an aggressively reduced q-ary tail is far from the LWE-derived d(LN) reference. Report the unsigned profile structure, never the signed advantage.

7 Re-Validating the fplll GSO Fix on a New Modulus Family

The companion paper documents a catastrophic-cancellation bug in fplll's squared-form GSO update at cryptographic moduli, and a Kahan-compensated fix. We re-validate that fix on a new (NTRU) modulus family: on $q \in \{971, 1087, 1201\}$ at $n = 127$, the four contaminated seeds recover from the precision-floor clamp ($b_1 = -345.388$) into the control band (≈ -0.1); the patched build passes all 15 fplll regression tests. The fix is thus validated on two modulus families (the companion paper's $q = 3329$ LWE, 38%$\to$0% degeneracy, and this NTRU set).

8 Discussion

Three observations bear on how SD-BKZ is modelled in NTRU security estimation. First, the SD-BKZ advantage is *not* a smooth function of dimension: the sharp onset spike at $n \approx 71$–73 (Section 4), exceeding the plateau, means a security estimate interpolated from a coarse n-grid can miss where self-dual reduction is most effective. Second, the quantity that captures the SD-vs-BKZ difference on NTRU is reference-free: the signed $d(\mathrm{LN})$ advantage flips sign under the reference and must not be reported (Section 2.3); the defensible observables are the unsigned profile divergence and the DSD flag. Third, and most consequential, the DSD-onset *gap*—SD-BKZ reaching dense-sublattice discovery at lower q than BKZ—grows with dimension at $\beta = 20$ (Section 5). At a matched $\beta = 40$ the enumeration and sieving oracles instead *agree*—both reach DSD-onset together, with no SD-vs-BKZ gap (Section 6); the gap is a $\beta = 20$ phenomenon that does not overlap the sieve-meaningful regime. Where the two oracles differ at $\beta = 40$ is the aggressiveness of sub-fatigue tail reduction—the G6K sieve clears the shortest GS vector markedly more than fplll enumeration on identical instances. A model that treats SD-BKZ and BKZ as RHF-equivalent (as the Root Hermite Factor, blind here, would suggest) misses that at $\beta = 20$ self-dual reduction crosses the overstretched threshold earlier.

These findings sit alongside two lines of work. The $\beta = 40$ regime and the Z-shaped q-ary profile geometry match the small-modulus forgery setting of Ducas–Espitau–Postlethwaite [DEP23], where the same profile structure is the attack surface. And the move away from an RHF-centric view toward sublattice density echoes recursive lattice reduction [AEPSD23], which abandons the Hermite-factor framing for a Rankin/dense-sublattice one—the same direction our reference-free observable points.

We are careful about scope. The dimension-onset spike is shown at $\beta = 20$, $q = 97$; the cross-engine comparison at $\beta = 40$, $n = 89$. The signed $d(\mathrm{LN})$ is reported nowhere as a result; and the cross-engine comparison at $\beta = 40$, $n = 89$ is an honest null on the DSD-onset (a regime constraint, not a confirmation), its positive content being the sieve-vs-enum tail-reduction difference. And $n = 127$ is excluded from the onset trend (Section 5): the fplll §8 cancellation contaminated those cells, and the circulant crack there is off-grid—it is a validation target (Section 7), not a trend point.

9 Reproducibility

Every result traces to per-seed JSON files under `results/seeds/`, SHA-256-gated for byte-identity and reproducible across machines from digest-pinned containers (Section 3.2). Each experiment is a named campaign in `config/sweep.toml`, so a cell is regenerated by a single `run_campaign` invocation naming the campaign, q, and seed count. The cross-engine soundness records and DSD-onset tables are committed under `results/validation/` with their own reproduction commands; the determinism and self-dual-semantics decisions are recorded in ADRs 005–008. The figures regenerate from the seed data through one figure module each, gated against drift. The companion paper's SHA baselines are preserved unchanged: the NTRU generator and the G6K engine were added without altering the seed JSON schema.

10 Conclusion

On NTRU, the difference between BKZ and self-dual BKZ that the Root Hermite Factor cannot see is real, reference-free, and consequential: SD-BKZ reaches dense-sublattice discovery at a lower modulus than BKZ, with a gap that grows with dimension at $\beta = 20$. At a matched $\beta = 40$ the enumeration and sieving oracles agree on the onset—there they differ instead in how aggressively they reduce the sub-fatigue tail, the sieve markedly more than enumeration. The companion paper argued the RHF is blind on LWE; this report shows the blindness persists on NTRU and on a sieve, exactly where it matters for the overstretched transition. The apparatus—a determinism-gated, containerised, two-engine benchmark with per-seed evidence—is released so the measurements can be reproduced and extended.

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